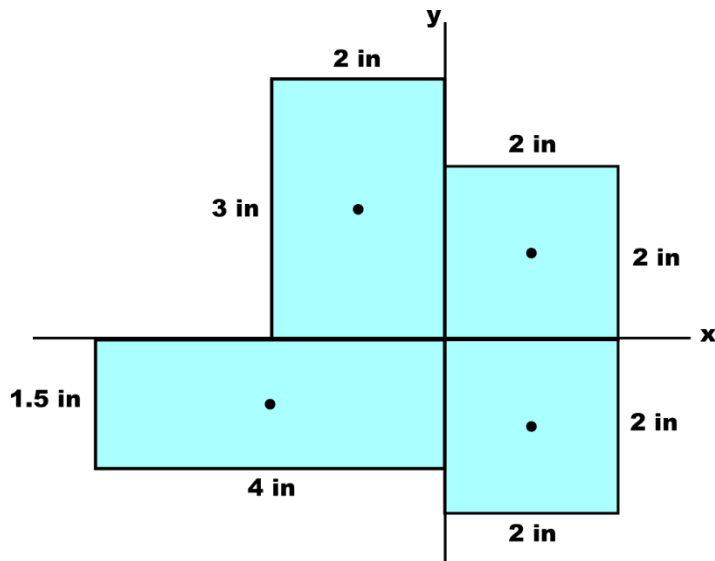


Ixy Problem which is not symmetric but that has Ixy = 0
(And thus Ix and Iy are max/min)



Problem which is not symmetric but yet Ixy = 0

Area						
No	Width	Height	Area	dx	dy	Adxdy
1	2.00	2.00	4.00	1.000	1.000	4.000
2	2.00	3.00	6.00	-1.000	1.500	-9.000
3	4.00	1.50	6.00	-2.000	-0.750	9.000
4	2.00	2.00	4.00	1.000	-1.000	-4.000
Ixy = Sum of Adxdy =						0.000

$$\begin{aligned}
 I_x &= \sum I_{base} \\
 &= \frac{1}{3}(2)(2)^3 + \frac{1}{3}(2)(3)^3 \\
 &\quad + \frac{1}{3}(4)(1.5)^3 + \frac{1}{3}(2)(2)^3 \\
 I_x &= \frac{1}{3} [2^5 + 2(27) + 4(1.5)^3] = \\
 &= 33.17 \text{ in}^4 = I_{MIN}
 \end{aligned}$$

$$\begin{aligned}
 I_y &= \sum I_{base} \\
 &= \left[\frac{1}{3}(2)(2)^3 \right] \times 2 + \frac{1}{3}(3)(2)^3 + \frac{1}{3}(1.5)(4)^3 = \frac{1}{3} [2^5 + 3(8) + 1.5(4)^3] \\
 I_y &= 50.67 \text{ in}^4 = I_{MAX}
 \end{aligned}$$